Dynamic Enteropancreatic Phenotyping via Sparse mFACEs and PTP/IEP Classification: An Ecological Meta-Analysis of Six Metabolic Cohorts

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Abstract

Context. Periprandial trajectories of the enteropancreatic axis (ghrelin, GIP, GLP-1, PYY, glucagon, insulin, glucose; 11 analyte-forms) undergo cohort-specific remodeling under obesity, type 2 diabetes (T2DM), their co-occurrence, and metabolic surgery. Quantitative multi-hormone integration of aggregate published data — capturing *physiologic periprandial responsiveness* across fasting and postprandial phases — remains statistically unresolved.

Objective. To establish six cohort-specific multi-hormone signatures via two routes — joint sparse mFACEs (Chiou-normalized) and per-analyte Periprandial Transition Profile (PTP) → Integrated Enteropancreatic Pattern (IEP) Type I–V — under a pre-registered protocol; glycemic subclassification (a/b/c) is restricted to dysphysiological types III, IV, V.

Design, Setting, and Participants. Ecological cohort-time-arm meta-analysis of 71 (Author × Cohort) tuples from 23 published source studies harmonized to six canonical cohorts: `no_obese_without_T2DM` (reference), Obesity, T2DM, Obesity_T2DM, SG, RYGBP. Pseudo-individualization via Gaussian process AR(1) draws (M = 1000; $\rho = 0.5$) yields N = 2750 pseudo-subjects across 55 arms.

Main Outcome Measures. Number K of retained joint mFPC components at FVE 0.90 with PC1 incretin loading; omnibus and pairwise Pillai F (descriptive cohort-separation magnitudes); cohort-specific prevalence of the joint 6-class scheme and of IEP Types I–V (with and without glycemic subtype); cross-framework Spearman concordance; classification-stage bootstrap stability vs. the ≥ 80 % pre-registered target.

Results. K = 14 components; PC1 incretin loading 0.817; N/K = 196. Omnibus Pillai F = 47.6 (reported as cohort-separation magnitude; permutation p saturated $< B^{-1} \approx 2 \times 10^{-4}$ under pseudo-IPD inflation). Pairwise F vs. reference: T2DM 108.3 > RYGBP 86.8 > Obesity_T2DM 59.3 > SG 34.1 > Obesity 22.5. IEP Type IV.II: RYGBP 76 % > Obesity_T2DM 56 % > SG 55 % > Obesity 45 % > T2DM 29 %. With glycemic subclassification, IV.II.c = 63 % (RYGBP) and 56 % (Obesity_T2DM). Dual-route Spearman $\rho = 0.50$ (complementary). Stability = 81.6 %.

Conclusions. Sparse mFACEs with Chiou normalization plus dual-track PTP/IEP classification is a reproducible framework for multi-hormone signatures from ecological aggregate data. It operationalizes *physiologic periprandial responsiveness* as a multi-analyte alternative to static *secretory capacity*. The RYGBP signature (IEP IV.II + Enhanced incretin) is distinguishable from T2DM (III.II infra-physiological), supporting functional stratification of bariatric-surgery candidates.

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1 Introduction

The convergent epidemics of obesity and type 2 diabetes mellitus (T2DM) constitute the dominant metabolic burden of the 21st century. The NCD Risk Factor Collaboration estimated that more than one billion individuals were living with obesity in 2022, with obesity now the prevailing form of malnutrition in 177 countries for women and 145 countries for men;[1] in parallel, 828 million adults had diabetes globally in 2022 — a fourfold increase from 198 million in 1990 — with 59 % remaining untreated.[2] The Global Burden of Disease 2023 analysis ranked diabetes as the third leading non-communicable disease worldwide and documented the largest age-standardized DALY rate increase among leading NCDs (+14.9 % since 2010).[3] Projections indicate that more than half of all adults — approximately 3.8 billion individuals — will have overweight or obesity by 2050,[4] while the International Diabetes Federation's 11th Diabetes Atlas projects 853 million adults with diabetes by 2050 and over one trillion dollars in annual health expenditure.[5] The recent Lancet Commission on Clinical Obesity, endorsed by more than 75 organizations, has further underscored the systemic nature of this crisis by proposing a redefinition of obesity that moves beyond body mass index to incorporate functional organ-specific assessments.[6]

Bariatric-metabolic surgery remains the most effective intervention for durable weight loss and T2DM remission in individuals with severe obesity. The 10-year follow-up of the Mingrone trial demonstrated T2DM remission rates of 25 % after Roux-en-Y gastric bypass (RYGBP) and 50 % after biliopancreatic diversion versus 5.5 % under medical therapy, with surgery reducing diabetes-related complications by 93 %.[7] The SM-BOSS and SLEEVEPASS 10-year randomized comparisons of sleeve gastrectomy (SG) versus RYGBP showed broadly comparable weight-loss durability with RYGBP exhibiting modest superiority on excess weight loss and T2DM remission.[8,9] The STAMPEDE 5-year outcomes confirmed the superiority of both procedures over intensive medical therapy for glycemic control,[10] and the recent ARMMS-T2D 12-year consortium data have extended these benefits to health-related quality of life;[11] a meta-analysis of nine randomized trials with $\geq$ 5-year follow-up corroborated RYGBP's superiority for T2DM remission while confirming SG's favorable gastroesophageal-reflux profile.[12] Crucially, the metabolic benefits of surgery extend far beyond caloric restriction and malabsorption, reflecting a profound remodeling of the enteropancreatic hormonal milieu: RYGBP produces marked amplification of postprandial GLP-1 and PYY secretion through accelerated nutrient delivery to distal L-cells,[13,14] while SG achieves durable ghrelin suppression through fundectomy-mediated removal of ghrelin-producing X/A-like cells.[15] These procedure-specific hormonal remodeling patterns create physiologically non-

equivalent enteropancreatic states that demand separate analytical treatment, and define metabolic phenotypes with greater fidelity than any static marker individually.[16-18]

The biological rationale for multi-hormone trajectory analysis is grounded in the incretin system's architecture. GLP-1 and glucose-dependent insulinotropic polypeptide (GIP) together account for 50–70 % of postprandial insulin secretion — the incretin effect — through complementary but non-redundant receptor-mediated mechanisms.[19] The recent resolution of the GIP paradox — demonstrating that GIPR agonism and antagonism decrease body weight through mechanistically distinct central nervous system pathways[20] — has fundamentally altered the interpretation of GIP trajectories in disease and post-surgical cohorts, shifting emphasis from absolute circulating levels to receptor-level and temporal signaling dynamics. The emerging multi-agonist pharmacological landscape — exemplified by tirzepatide's dual GIP/GLP-1 receptor co-agonism,[21] semaglutide's GLP-1 receptor agonism,[22] retatrutide's triple GLP-1/GIP/glucagon agonism,[23] CagriSema's amylin/GLP-1 dual targeting,[24] and survodutide's GLP-1/glucagon dual agonism in MASH[25] — further validates the principle that metabolic improvement depends on the coordinated modulation of multiple enteropancreatic axes, not isolated single-hormone targeting. Complementary discoveries in the 2024–2026 period have substantially enriched the biological context: gastric Piezo1 mechanosensitive channels regulate ghrelin synthesis and food intake,[26,27] the endogenous ghrelin-receptor antagonist LEAP2 modulates appetite homeostasis,[28] microbial DPP-4 isozymes degrade incretins in the gut lumen before absorption,[29] the GLP-1(9–36) metabolite retains independent vasodilatory and neuroprotective bioactivity,[30] and the liver–α-cell amino-acid axis reframes the interpretation of glucagon dynamics across disease and post-surgical states.[31]

1.1 Conceptual frame: physiologic periprandial responsiveness

A central conceptual choice of this work is to replace the conventional construct *secretory capacity* with *physiologic periprandial responsiveness* throughout the analytic framework. The two terms are not synonymous. *Secretory capacity* refers to the maximal biosynthetic output potential of an endocrine cell type — its ER folding kinetics, granule biogenesis, hormone-storage pool — and is operationally measured via maximal-stimulation tests. By contrast, *physiologic periprandial responsiveness* captures the stimulus-coupled dynamic behavior of the enteropancreatic system across both fasting and postprandial phases of the circadian cycle, including not only secretion but also the equally important *inhibition* phase: postprandial ghrelin suppression, fasting glucagon stabilization, periprandial insulin restraint between meals.[13,16,19] A construct that describes only secretion cannot describe what happens when the dominant biological event is inhibition; FPCA-decomposed periprandial trajectories from cohort-level mean curves capture both, motivating the deliberate neologism. The framework treats every analyte over the entire 0–180 min window relative to a meal, with explicit baseline (fasting) reference, so that

suppressive trajectories (e.g., ghrelin under satiety) and secretory trajectories (e.g., postprandial insulin) receive equivalent functional decomposition.

This responsiveness is shaped by a multidimensional set of variables that no single phenotype captures: age, sex, energy balance, BMI, dysmetabolic conditions of genetic or acquired origin (Prader-Willi syndrome, T1DM, T2DM, prediabetes, overweight, obesity, anorexia nervosa), gastrointestinal tract integrity or its surgical/endoluminal modification, the neuroplasticity state of the enteropancreatic system, and intake-related variables (liquid vs. solid, caloric load, ingestion velocity, gastric emptying rate, macronutrient proportions, and prior intake history).[6,18] Together these variables generate a multidimensional, dynamic, and heterogeneous endocrine response. The framework deliberately accepts within-cohort heterogeneity as an expected epiphenomenon — reflecting differences in caloric protocols, sampling schedules, assays, units, and patient-level variables across source studies — rather than as a methodological failure. Multi-level standardization across this irreducible heterogeneity is itself part of the contribution.

1.2 Statistical foundations

Sparse functional data analysis and functional principal component analysis[32] — specifically the sparse PACE algorithm of Yao, Müller and Wang,[33] its multivariate extension by Happ and Greven,[34] and the *mixed-effects fast covariance estimation* of Li, Wang, Liu and Greven[35] — provide the natural methodological framework to decompose irregularly sampled periprandial trajectories into orthogonal modes of variation (global amplitude PC1, early-vs-late redistribution PC2, higher-order shape PC3). The Chiou, Chen and Yang normalization[36] preserves balanced per-hormone contribution in the joint decomposition by equalizing integrated variances, operationally indispensable when hormone scales differ across three orders of magnitude (ghrelin ~200 pmol/L vs. GLP-1 active ~5 pmol/L). Complementarily, the PTP/IEP classification framework (v1.0, April 2026) formalizes cohort-time-arm taxonomy through six *Periprandial Transition Profiles* per analyte (Preserved, Borderline Impaired, Impaired, Blunted, Borderline Altered, Altered — plus three post-intervention classes Recovered, Borderline Enhanced, Enhanced) and five *Integrated Enteropancreatic Pattern Types* (I–V), constituting a deterministic biological taxonomy complementary to the multivariate inferential analysis.

1.3 Aim

Here we apply both frameworks in integrated form to an aggregated *master data table* of 71 cohort-time-arm tuples from 23 published source studies[49-71] covering 11 analyte-forms. The operational objective is to establish and statistically validate six cohort-specific multi-hormone signatures — reference `no_obese_without_T2DM`, `Obesity` without T2DM, `T2DM` without obesity,

co-occurring `Obesity_T2DM`, post-`SG`, post-`RYGBP` — under a fully pre-registered protocol (OSF DOI 10.17605/OSF.IO/3CZRE), with companion analysis code archived at Zenodo (concept DOI 10.5281/zenodo.19743544; current version v1.3 10.5281/zenodo.19758429). Per the pre-registered scope of this study, the IEP Type taxonomy is presented in two parallel tracks — without and with glycemic subclassification — to deepen the analysis and to highlight the added value of glycemic subphenotyping. Glycemic subclassification (a/b/c) is restricted to Types III, IV, and V; Patterns I and II by definition imply preserved or recovered glycemic regulation (subtype a) and admit no other glycemic subtype. The study is reported in accordance with the STROBE cohort checklist;[46] TRIPOD+AI[47] is not applicable because the analysis is descriptive-inferential rather than predictive at the individual level.

2 Methods

2.1 2.1 Design, data source, and inferential scope

Ecological meta-analysis at the cohort-time-arm level following the pre-registered v10.0 master analysis plan (OSF DOI 10.17605/OSF.IO/3CZRE; section references below cite specific paragraphs of that plan). The *master data table* (input SHA-256 `ce2e343d...`) consolidates 71 unique (Author × Cohort) tuples from 23 independent source publications. Because all data derive from published aggregate summaries rather than individual participant records, the primary unit of analysis is the cohort-time-arm — defined as a unique combination of source study, clinical cohort, intervention status, and post-intervention time point. The inferential unit is declared explicitly for every result: cohort-time-arm level for FPCA-derived descriptors, and source-reported scalar-summary level for fasting or AUC values extracted directly from publications.

2.2 2.2 Cohort definitions

Forty-seven heterogeneous source labels were mapped to six canonical cohorts via pre-registered normalization rules (decisions D1–D4 in the preregistration YAML):

- D1. Pre-surgery cohorts pooled into `Obesity` or `Obesity_T2DM` with `surgery_status` metadata.
- D2. Post-bariatric T2DM status captured as `had_t2dm_pre_surgery` $\in$ {`TRUE`, `FALSE`, `unknown`}.
- D3. Six caloric-restriction labels pooled into `Obesity` with a `caloric_restriction_post` tag retained in a sensitivity column.
- D4. The "Healthy Middle-aged" obesity descriptor does not define a distinct cohort and is pooled into `Obesity`.

The six canonical cohorts are: `no_obese_without_T2DM` (reference; non-obese, non-diabetic, no metabolic-surgery history), `Obesity` (BMI $\geq$ 30 kg/m^2 without T2DM or IGT), `T2DM` (diagnosed T2DM per source criteria, without concurrent obesity), `Obesity_T2DM` (co-occurring obesity and T2DM), `SG` (post-sleeve gastrectomy at specified post-operative time points), and `RYGBP` (post-Roux-en-Y gastric bypass at specified time points). SG and RYGBP are treated as physiologically non-equivalent procedures and never pooled.

2.3 2.3 Analyte panel and biological-role classification

The eleven analyte-forms are organized by biological dominance phase, complementary to the framework-level *secretion-dominant / suppression-dominant / glycemic-regulation* groupings of PTP framework v1.0 §2.2:

- Fasting hormones (active during the inter-meal/fasting state, with key inhibition events at meal onset): ghrelin (total and acylated) and glucagon. Ghrelin exerts critical biological functions during fasting — maintaining neuronal plasticity, inhibiting fasting insulin secretion, and modulating appetite and anxiety circuits via the GHSR axis;[27,28] chronic post-bariatric ghrelin reduction has been observationally associated in the clinical literature with neurobehavioral and addiction-related sequelae warranting independent investigation. Glucagon stabilizes fasting glycemia via hepatic, pancreatic, and peripheral glucoregulatory effects, with the recently described liver–α-cell amino-acid axis providing additional regulatory input.[31]
- Postprandial hormones (active during the post-meal nutrient-stimulated phase): GIP (total and active), GLP-1 (total and active), PYY (total and 3-36), and insulin. These hormones drive nutrient-stimulated insulin secretion, satiety signaling, and glycemic excursion control.[16,19]
- Glycemic regulation signal: glucose. Not a secretory hormone but the regulated variable whose periprandial trajectory reflects the integrated activity of all of the above.

Glycemia is regulated during fasting by ghrelin-mediated insulin inhibition plus glucagon-driven glucoregulation, and post-prandially by GLP-1- and GIP-driven insulin stimulation; the role of glucagon in postprandial hyperglycemia is context-dependent (e.g., elevated in T2DM, suppressed at appropriate post-bariatric checkpoints to prevent reactive hypoglycemia). The framework explicitly evaluates both states because their proper characterization requires inclusion of fasting baselines and not only postprandial dynamics.

The basal (fasting) value B_{ih} is therefore included as the first coordinate of the per-analyte PTP classification vector $Z_{ih} = [z(B_{ih}), z(\xi_{ih1}), z(\xi_{ih2}), z(\xi_{ih3})]$. Omitting it would discard the ability to identify

periprandial changes that originate in the fasting state (e.g., elevated fasting GIP that precedes islet dysfunction and dysinsulinemia even before prediabetes onset; reduced fasting ghrelin in obesity inversely correlated with BMI; recovered fasting ghrelin in long-term post-RYGBP patients with sustained weight loss).

Total and active forms of GIP and GLP-1 were analyzed as separate analyte-form strata because DPP-4 processing differentially inactivates GIP and GLP-1 while activating PYY(1–36) to PYY(3–36); the GLP-1(9–36) metabolite retains independent vasodilatory and neuroprotective bioactivity,[30] and microbial DPP-4 isozymes that degrade incretins in the gut lumen before absorption[29] provide additional mechanistic justification for distinguishing total from active forms.

2.4 2.4 Challenge-class handling and time alignment

Three challenge classes are represented in the *master data table*: solid mixed-meal test (SMMT), liquid mixed-meal test (LMMT), and oral glucose tolerance test (OGTT). The common analysis window is defined as [0, 180] minutes, with all trajectories aligned to $t = 0$ at stimulus ingestion. Time points are mapped to a common 1-min evaluation grid via piecewise cubic Hermite interpolating polynomial (PCHIP) interpolation, which preserves monotonicity and prevents overshoot artifacts at physiological boundaries (e.g., ghrelin nadir, glucose return to baseline). Basal concentration B_{ih} is defined as the $t = 0$ value for the reference cohort and the nearest-to-zero value for the study cohorts. Where within-stratum sample sizes are insufficient, SMMT and LMMT are combined into a "mixed meal" super-stratum with a binary covariate encoding stimulus form; OGTT is analyzed separately.

2.5 2.5 Unit harmonization and data extraction

All peptide hormones are expressed in pmol/L and glucose in mmol/L (SI units), with conventional units reported parenthetically per JCEM convention. Conversion factors apply standard molecular weights when source studies report in pg/mL, μU/mL, or mg/dL. SEM is converted to SD via $\text{SD} = \text{SEM} \times \sqrt{n}$ where n is the reported sample size. Time-course data not available in tables were extracted from published figures by the sole author using WebPlotDigitizer v4.7. Tabular data take priority over digitized values per a strict source-evidence hierarchy (canonical CSV *master data table* → extracted numerical tables → digitized values).

2.6 2.6 Pseudo-individualization and variance priors

The *master data table* reports cohort means without SD/SEM; trajectory variance is estimated via pseudo-IPD generation as specified in the pre-registration (v10.0 §4.2 — "Pseudo-individualization

via Gaussian process draws"). For each (Author, cohort, hormone), M = 1000 synthetic trajectories are sampled from a Gaussian process $\mathrm{GP}(\mu_{\mathrm{cohort}}(t), K_\rho(s,t))$ with continuous-time AR(1) kernel $K_\rho(s,t) = \rho^{|s-t|/30\,\mathrm{min}}$ and marginal variance $\sigma_h^2(t) = (\mathrm{CV}_h \times \mu_h(t))^2$, where CV_h is the hormone-specific coefficient of variation drawn from literature (upper-bound typical values: ghrelin 0.40, GLP-1 total 0.35, GIP total 0.30, insulin 0.20, glucose 0.12; full table in the preregistration YAML). The primary run uses $\rho = 0.5$ with CV multiplier 1.0; sensitivity analyses cover $\rho \in \{0.3, 0.7, 0.9\}$ and CV multipliers $\{0.75, 1.25\}$. A deterministic subsample of N = 50 pseudo-subjects per arm (2750 total across 55 productive arms) is used for the mFACEs joint fit; the full M = 1000 archive is deposited on Zenodo.

2.7 2.7 Multivariate FPCA — joint mFACEs (primary)

This approach adapts the longitudinal-FPCA framework of Greven and colleagues.[38] It employs per-analyte FACE fits via `face::face.sparse`.[39] We estimate pairwise cross-covariance blocks using 2-D tensor-product penalized splines.[35] Explicit Chiou normalization[36] is then applied during block assembly, operationalized analogously to the group-specific functional subspace construction of Schmutz and colleagues.[40] BLUP score recovery retains measurement-error variance σ^2 in the observation model.[38] Adaptive ridge regularization is applied. Joint eigendecomposition yields K retained components satisfying FVE ≥ 0.90 (primary) or 0.95 (sensitivity); the FVE-threshold rationale follows the robust-FPCA component-retention framework of Wang and colleagues.[41] Identifiability requires $N/K_{\mathrm{eff}} > 10$, a rule-of-thumb consistent with the multivariate FPCA review of Wang, Chiou and Müller[32] and supported by classical asymptotic results in single-trajectory FPCA.[42] Sign alignment of eigenfunctions across bootstrap replicates is enforced via `flipFuns()` with Procrustes rotation.[43]

2.8 2.8 Univariate FPCA (sensitivity)

Each of 11 analyte-forms is modeled separately by PACE on the dynamic component $d_{ih}(t) = x_{ih}(t) - B_{ih}$ (pre-registration v10.0 §7.1 — "univariate PACE on dynamic component"), with baseline B_{ih} defined as the strict $t = 0$ value for the reference cohort and the nearest-to-zero value for the study cohorts. Minimum 3 observations per subject per hormone; FVE 0.95. The semantic structure followed: PC1 = global amplitude; PC2 = early-versus-late temporal redistribution (phase/timing shift); PC3 = biphasic or higher-order temporal shape. Components that cannot be stably oriented physiologically are retained numerically but the physiologic label is withheld.

2.9 2.9 Inference and dual classification

Formal inference by FANOVA permutation (B = 5000 cohort-label permutations per retained PC; BH-FDR $\alpha = 0.05$) plus Pillai multivariate contrasts. Simultaneous 95 % bands via sup-t bootstrap (B = 2000),[48] implemented analogously to the corrected functional confidence-band construction of Goldsmith and colleagues.[37] Dual classification:

- Joint-mFPC 6-class on the first three retained multivariate scores (ξ_1, ξ_2, ξ_3): pseudo-subjects z-standardized against the reference distribution, then assigned to one of six classes (Preservado, Impairment limítrofe, Impaired, Blunted, Enhanced, Altered) with pre-registered precedence Altered > Blunted/Enhanced > Impaired > Impairment limítrofe > Preservado.
- Per-analyte PTP → IEP Type I–V (framework v1.0): classification vector Z_{ih} from univariate PACE fits, six Primary PTP classes assigned with pathophysiological filter (Altered restricted to GIP, insulin, glucose, ghrelin; upper-tail GLP-1/PYY/glucagon routed to Enhanced/Borderline Enhanced when glucose subtype = a). Cohort-time-arm IEP Types assigned by the deterministic 8-rule precedence hierarchy specified in §2.10.

2.10 2.10 Deterministic PTP/IEP construction

The PTP/IEP framework v1.0 (April 2026) was developed by the author and is reproduced here in full as the normative specification used to generate every IEP assignment in the Results section (full framework document deposited as Supplementary File S4). Because the framework was both designed and applied to the present ecological corpus, all IEP-track findings should be regarded as confirmatory applications of the pre-registered scheme rather than as independent validation; external testing in independent IPD cohorts is identified as a priority next step (§4 Limitations). The framework distinguishes (a) six primary Periprandial Transition Profiles assigned per analyte-form to disease-state cohorts, (b) three additional secondary (post-intervention) PTPs, (c) five Integrated Enteropancreatic Pattern Types (I–V) with .I/.II refinement subtypes, and (d) three glycemic subtypes (a/b/c) restricted to Types III, IV, V. Classification proceeds by computing the standardized analyte-specific vector $Z_{ih} = [z(B_{ih}), z(\xi_{ih1}), z(\xi_{ih2}), z(\xi_{ih3})]$ and applying the percentile-fallback operational rules.

2.10.1 2.10.1 Primary PTP categories (cross-sectional disease characterization)

PTP category	Operational percentile fallback (basal / PC1)	Physiological meaning
Preserved	Within central reference band P25–P75; no primary coordinate crosses tail abnormality	Reference-concordant physiologic periprandial responsiveness
Borderline Impaired	P10–P25 (mild lower-tail); no primary coordinate < P5	Mild reduction in physiologic periprandial responsiveness
Impaired	P5–P10 (moderate lower-tail); no primary coordinate < P5	Moderately reduced responsiveness
Blunted	< P5 (severe lower-tail)	Severely attenuated responsiveness
Borderline Altered	P75–P95 (mild upper-tail in pathophysiologically relevant direction)	Mildly abnormal upper-tail dysregulation
Altered	≥ P95 (extreme upper-tail)	Marked dysregulation; reserved for excessive GIP, hyperinsulinemic patterns, hyperglycemic excursion, and context-specific ghrelin excess

Upper-tail classes (Borderline Altered, Altered) are applied only when the excess is pathophysiologically meaningful for that analyte in that context: high GIP in cross-sectional disease cohorts is not automatically classified as Altered without clear biological justification.

2.10.2 2.10.2 Secondary PTPs (post-intervention)

For post-intervention arms (caloric restriction, bariatric surgery, pharmacotherapy), three additional PTPs apply: Recovered (previously abnormal profile shifts into the reference-concordant region; explicit pre→post improvement required), Borderline Enhanced (mild physiologically coherent supra-reference increase with preserved/recovered glucose; reserved primarily for GLP-1, PYY, and contextually insulin), and Enhanced (marked physiologically coherent supra-reference increase, reserved mainly for GLP-1, PYY, and carefully selected insulin responses with preserved glycemic regulation). Enhanced is not assigned if the accompanying glucose pattern is altered or dysglycemic; such profiles are classified as Altered. Weight loss is treated as a contextual modifier, not a hard gate.

2.10.3 2.10.3 Non-glucose PTP state groups

For integrated typing, non-glucose analyte-level PTPs are collapsed into the following ordered groups:

- $R = \{\text{Preserved, Recovered}\}$
- $L_1 = \{\text{Borderline Impaired}\}$
- $L_2 = \{\text{Impaired, Blunted}\}$
- $U_1 = \{\text{Borderline Enhanced}\}$
- $U_2 = \{\text{Enhanced}\}$
- $D_1 = \{\text{Borderline Altered}\}$
- $D_2 = \{\text{Altered}\}$

2.10.4 2.10.4 Glucose subtype groups

Glucose is treated as a glycemic regulation signal rather than as an enteropancreatic hormone for integrated assignment, and its PTP is collapsed into three glycemic subtypes:

- $a = \{\text{Preserved, Recovered}\}$
- $b = \{\text{Borderline Impaired, Impaired, Blunted}\}$
- $c = \{\text{Borderline Altered, Altered}\}$

2.10.5 2.10.5 Classifiability rule

An integrated pattern is assigned only when (1) glucose PTP is classifiable, (2) at least two non-glucose analyte-level PTPs are classifiable, and (3) the classifiable non-glucose set includes at least one pancreatic effector signal (insulin) and at least one gut-hormone signal (ghrelin, GIP, or GLP-1). Cohort-time-arms not meeting these criteria are classified as not integrable; missing analytes are not imputed.

2.10.6 2.10.6 Deterministic precedence hierarchy (8 rules)

Integrated pattern assignment follows the strict 8-rule deterministic precedence hierarchy (rules evaluated in order; first satisfying rule terminates assignment):

1. If any non-glucose analyte was Altered → Type IV.II.
2. Else if any non-glucose analyte was Borderline Altered → Type IV.I.
3. Else if any non-glucose analyte was Enhanced:
    a. if glucose subtype = a and no non-glucose analyte was in L_1 or L_2 → Type II.II;

   b. otherwise → Type V.II.

4. Else if any non-glucose analyte was Borderline Enhanced:

   a. if glucose subtype = a and no non-glucose analyte was in L_1 or L_2 → Type II.I;
   b. otherwise → Type V.I.

5. Else if any non-glucose analyte was Impaired or Blunted → Type III.II.
6. Else if any non-glucose analyte was Borderline Impaired → Type III.I.
7. Else if all classifiable non-glucose analytes were Preserved → Type I.I.
8. Else if at least one non-glucose analyte was Recovered and all remaining classifiable non-glucose analytes were Preserved or Recovered → Type I.II.

All five integrated types (Type I Physiological; Type II Supra-physiological coherent; Type III Infra-physiological; Type IV Dysphysiological; Type V Supra-physiological with residual dysregulation) are then subtyped by glycemic status (a/b/c) for Types III, IV, V; Patterns I and II by definition imply preserved or recovered glycemic regulation (subtype a) and admit no other glycemic subtype.

2.10.7 2.10.7 PTP/IEP classification scope and glycemic subclassification

The IEP framework distinguishes five integrated patterns (Type I physiological, Type II supra-physiological coherent, Type III infra-physiological, Type IV dysphysiological, Type V supra-physiological with residual dysregulation) with refinement subtypes (.I/.II reflecting borderline vs marked deviation). Glycemic subclassification a/b/c is restricted to Types III, IV, and V, where dysregulation of the enteropancreatic axis can plausibly couple to glycemic dysregulation. Patterns I and II by definition exclude dysglycemic phenomena: a Type I.I/I.II profile is reference-concordant or restored across all classifiable non-glucose analytes, and a Type II.I/II.II profile is supra-physiological coherent with preserved glucose by gating rule. Therefore the origin of glucose alterations in Patterns I and II would not be secondary to enteropancreatic dysregulation, and glycemic subtypes are not applied to these patterns. This restriction is conceptually consistent with the framework's grounding on enteropancreatic secretory dynamics and acknowledges as a partial weakness that dynamic secretion is not always proportional to biological action capacity (Type II as the ideal therapeutic restoration target embodies this distinction precisely).

For Patterns III, IV, and V the three subtypes are: subtype a glucose preserved or recovered; subtype b glucose with borderline impairment, impairment, or blunting (lower-tail); subtype c glucose with borderline alteration or alteration (upper-tail dysregulation). Per the pre-registered scope of this study, results are presented in two parallel tracks: (i) IEP Types I–V without glycemic subclassification — the integrated enteropancreatic pattern alone, deepening insight into the enteropancreatic axis irrespective

of the glycemic outcome; and (ii) IEP Types with glycemic subclassification on III/IV/V only — adding the glycemic subphenotype as a complementary layer.

2.10.8 2.10.8 Statistical and reporting rules

Five reporting rules govern the integrated typing output: (i) integrated typing is reported only in the classifiable subset; (ii) exact frequency estimates are interpreted as ecological cohort-time-arm counts, not participant-level prevalence; (iii) primary-corpus Type I–V frequencies are reported separately from corroborative source-reported pattern information; (iv) the secondary orthogonal shape-balance descriptor derived from ξ_2 and ξ_3 may be reported descriptively but does not replace ξ_2 and ξ_3 in integrated typing; and (v) when contradictory non-glucose PTP states are present, the precedence hierarchy in §2.10.6 determines the final integrated type.

2.11 2.11 Borderline cutoff design and reference cohort selection

The PTP percentile thresholds use borderline bands at P10–P25 (lower-tail Borderline Impaired) and P75–P95 (upper-tail Borderline Altered), with strict-tail flags at <P5 (Blunted) and ≥P95 (Altered). The choice of borderline bands rather than strict P25/P75 quartile boundaries with binary outcomes increases sensitivity to mild deviations and avoids dichotomizing subjects whose periprandial responsiveness lies near the reference periphery. This design parallels the diagnostic logic that medicine uses with imperfect statistical anchors (BMI as a population statistic; glucose as a consequence of beta-cell mass loss, insulin resistance, and enteropancreatic defects — neither captures the underlying pathophysiology). Diagnostic frameworks anchored to statistics (BMI cutoffs) or downstream consequences (glucose thresholds) yield pathologies that resist remission; frameworks anchored to underlying mechanisms (e.g., bariatric surgery's enteropancreatic remodeling) yield true remission. The pursuit of Pattern Type II (supra-physiological enteropancreatic responsiveness with preserved glycemia) as the ideal restoration target embodies this mechanism-anchored philosophy.

The reference cohort `no_obese_without_T2DM` (non-obese, non-diabetic) is selected because (i) intervention cohorts cannot serve as their own physiological reference — they are by definition pathological in the pre-intervention state; (ii) comparing post-intervention cohorts against the reference establishes the therapeutic value of the intervention more accurately than pre-vs-post pairing alone; and (iii) weight is determinant for discriminating intervention effects on PTPs, especially for ghrelin where fasting baseline and postprandial suppression both correlate inversely with BMI and with weight loss. The framework retains capacity for future expansion to secondary PTPs (paired pre-vs-post within cohort) and to additional intervention modalities (medications, dietary interventions, vagal stimulation, endoluminal procedures, gastric plication) that act on gastric, duodenal, or small-bowel regions through mechanisms distinct from resective bariatric surgery.

2.12 2.12 Validation and sensitivity analyses

Classification-stage bootstrap (B = 2000): scores from the primary fit held fixed; reference cohort resampled with replacement to perturb the z-score reference distribution; modal class per pseudo-subject + stability = Pr(modal). Pre-registered target: $\geq$ 80 % of subjects with stability $\geq$ 0.80. Pipeline-stage bootstrap (B = 50 primary + B = 300 extended Zenodo archive): full re-fit on resampled pseudo-IPD. Sensitivity matrices across ρ and CV multipliers report K, incretin loading, and qualitative cohort ordering. Pre-specified sensitivity analyses include: (i) PTP-threshold displacement by $±$0.5 SD with reporting of the proportion of assignments that change; (ii) leave-one-axis-out integrated typing repeated excluding each analyte in turn; (iii) ecological-robustness pseudo-IPD generated under AR(1) within-cohort temporal correlations $\rho \in \{0.3, 0.5, 0.7, 0.9\}$; and (iv) challenge-class pooling — comparison of stratum-specific vs. pooled SMMT+LMMT results with a binary covariate.

2.13 2.13 Reproducibility and computational environment

R 4.5.3 arm64 on Apple M4 Pro (macOS Sequoia 15), BLAS via Apple Accelerate; parallelism via `future::plan(multisession, workers = 8)`. Key packages: `fdapace` 0.6.0, `face` 0.1-7, `MFPCA` 1.3-11, `funData` 1.3-9, `conformalInference.fd` 1.1.1, `roahd` 1.4.3, `mgcv` 1.9, `pointblank` 0.12, `irr` 0.84. Seed 20260422 across all stochastic procedures. Random seeds were set and recorded for all stochastic operations (MCMC sampling, bootstrap resampling, pseudo-IPD generation). The analytical pipeline was pre-registered on OSF using the Secondary Data Preregistration template (DOI 10.17605/OSF.IO/3CZRE). Reporting adhered to a composite checklist: STROBE[46] for observational design elements, TRIPOD+AI[47] Items 16 (full model specification) and 23 (open code), and FDA-specific items: smoothing method, basis type, regularization parameter selection, number and interpretation of FPCs, and software versions. `renv.lock` and session info are deposited on Zenodo.

3 Results

3.1 Cohorts and data coverage

After pre-registered normalization, the ETL produced 2917 long-format observations across six canonical cohorts (Table 1, Figure 1). Hormone coverage is asymmetric (glucose 508 obs, GIP active 96 obs), reflecting source-study reporting heterogeneity rather than pipeline bias.

3.2 Pseudo-IPD and primary mFACEs fit

M = 1000 pseudo-subjects per arm; deterministic subsampling to N = 50 per arm yielded 2750 pseudo-subjects in 55 productive arms. The joint mFACEs fit with explicit Chiou normalization retained K = 14 components at FVE 0.90 (N/K = 196, §7.3 criterion cleared with large margin). FVE top-5 = {0.16, 0.28, 0.38, 0.48, 0.55}; without Chiou, PC1 alone captured 84 % of variance via ghrelin/insulin dominance and the incretin loading collapsed to 0.000 — confirming the operational necessity of explicit Chiou normalization. Eighty-nine percent of off-diagonal cross-covariance blocks were estimated; zero BLUP failures across 2750 pseudo-subjects.

3.3 Incretin axis and joint-mFPC classification

The argmax rule identified PC1 as the incretin axis with integrated squared loading 0.817 (PC2: 0.699, PC3: 0.608). The joint 3-PC classifier (Table 2) yields biologically coherent cohort-specific prevalence patterns (Figure 2A): reference 41.7 % Preservado dominant; RYGBP 37 % Altered + 12 % Enhanced (the only cohort with both extremes); T2DM 28 % Impaired + 9 % Blunted (incretin deficiency signature); Obesity_T2DM 28 % Impaired + 20 % Blunted (convergent dysregulation); SG 25 % Preservado with 5 % Enhanced; Obesity 34 % Preservado with moderate Impaired representation.

3.4 Formal inference

Omnibus Pillai test across 14 retained components produced F = 47.6. Pairwise Pillai contrasts vs. reference: T2DM (F = 108.3) > RYGBP (86.8) > Obesity_T2DM (59.3) > SG (34.1) > Obesity (22.5) (Table 3 Panel A, Figure 2B). Because pseudo-IPD generation inflates effective N (2750) beyond the underlying 71 (Author × Cohort) tuples, the empirical permutation distribution saturates against the minimum permutation grain $B^{-1} \approx 2 \times 10^{-4}$ for every contrast; classical inference under the F-distribution is therefore not warranted, and F values are reported as comparative cohort-separation magnitudes, not as small-sample test statistics. Per-PC analysis (Table 3 Panel B): RYGBP attains F = 496 on PC1 and F = 275 on PC7; T2DM peaks on PC7 (F = 370); PC4 is specifically salient for RYGBP (F = 244). PC7 emerges as a metabolic–diabetic disruption axis with high coverage in T2DM and Obesity_T2DM.

3.5 Trajectory differences and simultaneous bands

Of 47 hormone × cohort-vs-reference comparisons, 18 were significant at 95 % simultaneous coverage across the entire 0–180 min grid. Top maximum-difference findings (Figure 3): ghrelin total SG $\Delta = -143$ pmol/L (fundectomy signature); insulin Obesity $\Delta = +975$ pmol/L; insulin RYGBP $\Delta = +665$

pmol/L (exaggerated post-surgical response); glucose Obesity_T2DM $\Delta = +8.3$ mmol/L; ghrelin acyl SG $\Delta = -97$ pmol/L; GLP-1 total T2DM $\Delta = -12.8$ pmol/L (incretin deficiency).

3.6 PTP/IEP Type I–V — without glycemic subclassification

The pre-registered primary track presents IEP Type I–V distributions without glycemic subclassification (Table 4A, Figure 2C), interpreting the integrated enteropancreatic pattern alone independently of the glycemic outcome. Across the classifiable subset (1600 / 2750 = 58 %), the distribution by cohort is:

- `no_obese_without_T2DM` (reference): I.I = 6.8 %; III.I = 12.8 %; III.II = 12.5 %; IV.I = 47.8 %; IV.II = 18.8 %; V.I = 1.2 %; V.II = 0.2 %.
- Obesity: I.I = 2.9 %; III.I + III.II = 7.4 %; IV.I = 44.6 %; IV.II = 45.1 %.
- T2DM: III.II = 37.0 %; IV.I = 34.5 %; IV.II = 28.5 %.
- Obesity_T2DM: III.I + III.II = 10.4 %; IV.I = 33.7 %; IV.II = 56.0 %.
- SG: III.I + III.II = 2.0 %; IV.I = 43.0 %; IV.II = 55.0 %.
- RYGBP: III.I + III.II = 1.2 %; IV.I = 22.4 %; IV.II = 76.4 %.

The Type IV.II (dysphysiological marked) ranking — RYGBP 76 % > Obesity_T2DM 56 % > SG 55 % > Obesity 45 % > T2DM 29 % > reference 19 % — identifies multi-analyte enteropancreatic dysregulation strongest in post-bariatric and metabolic-disease cohorts, reflecting per-analyte upper-tail patterns dominated by GIP (in obesity contexts) and ghrelin/insulin (in surgical contexts). The 18.8 % Type IV.II prevalence in the reference cohort represents the residual arithmetic floor of the per-analyte multiple-testing logic noted in §2.10.6, mitigated but not eliminated by the pathophysiological filter; cohort-vs-reference comparisons should therefore be interpreted as relative excess prevalence (e.g., RYGBP–reference $\Delta = +57.6$ pp; Obesity_T2DM–reference $\Delta = +37.2$ pp) rather than as absolute disease prevalences. Pattern Type II (supra-physiological coherent) is undetectable in the present ecological corpus (< 0.5 % across all cohorts in both tracks). Two non-mutually-exclusive interpretations follow: (i) the available source studies undersample optimally-restored post-bariatric phenotypes (the corpus is dominated by within-1-year post-operative arms in cohorts with persistent glycemic dysregulation), and (ii) the Type II gating rule (glucose subtype = a strictly required) is conservative by construction; both interpretations identify Type II detectability as a target for prospective IPD studies in long-term remission cohorts. Pattern I.I (Physiological) reaches 6.8 % of classifiable reference subjects.

3.7 PTP/IEP Type I–V — with glycemic subclassification on III, IV, V

Restricting glycemic subclassification a/b/c to Types III, IV, and V (per the framework's scope and the conceptual restriction that I and II cannot couple to dysglycemia) yields the refined Table 4B distribution. The dominant cells are:

- IV.II.c (dysphysiological marked + dysglycemic): RYGBP 62.8 % > Obesity_T2DM 56.0 % > SG 47.0 % > Obesity 36.6 % > T2DM 28.5 % > reference 8.8 %.
- IV.II.a (dysphysiological marked + glucose preserved): RYGBP 10.4 %, SG 8.0 %, reference 7.8 %, Obesity 1.1 %.
- IV.II.b (dysphysiological marked + glucose impaired): Obesity 7.4 %, RYGBP 3.2 %, reference 2.2 %.
- IV.I.c (dysphysiological borderline + dysglycemic): Obesity 42 %, SG 39 %, RYGBP 18.8 %, reference 22.2 %.

Glycemic subphenotyping reveals that the dominant dysphysiological signature in metabolic-disease cohorts (Obesity_T2DM, RYGBP, T2DM) is overwhelmingly coupled to dysglycemia (subtype c), confirming the expected coupling. SG and Obesity show a more mixed profile across subtypes, with non-trivial subtype a (preserved glucose despite dysphysiology) suggesting sub-clinical enteropancreatic remodeling preceding overt dysglycemia. The reference cohort, while predominantly classified as IV.I/IV.II by the per-analyte multiple-comparisons arithmetic (see Discussion), shows an even glycemic-subtype distribution consistent with its clinical normalcy. This split provides incremental diagnostic and prognostic resolution that the unsubclassified Type I–V table cannot.

3.8 Cross-framework concordance

The Spearman rank correlation between joint Pillai F (cohort vs. reference) and IEP Type IV.II prevalence (without subclassification, 5 non-reference cohorts) is $\rho = 0.50$, a moderate concordance rather than near-redundancy (Figure 2D). The two rails identify the same extreme cohorts but with interpretable rank discordance: T2DM ranks #1 on Pillai F but #5 on Type IV.II prevalence because its signal is dominated by infra-physiological Type III.II (37 %) rather than dysphysiological Type IV.II — consistent with diabetic incretin deficiency where the dominant deviation is reduced secretion, not exaggerated response. Obesity_T2DM, by contrast, shows convergent Type IV.II prevalence at 56 % alongside moderate Pillai F (59), reflecting consistent multi-analyte upper-tail deviation across the full retained subspace.

3.9 Stability and robustness

Classification-stage bootstrap (B = 2000) yielded 81.6 % of pseudo-subjects with modal-class stability $\geq$ 0.80, exceeding the pre-registered 80 % target. Pipeline-stage bootstrap (B = 50): 50/50 reps converged; K median = 14 (range 13–15); incretin axis PC1 invariant; loading IQR 0.795–0.829. Extended Zenodo archive at B = 300: 0.29 pp median drift vs. B = 50 across 36 cohort $\times$ class pairs (IQR ratio 1.02),

empirically confirming bootstrap variance saturation at $B = 50$. Sensitivity: $\rho \in \{0.3, 0.7, 0.9\}$ preserves $K \in [11, 14]$; CV multipliers $\{0.75, 1.25\}$ preserve $K \in [12, 16]$ and incretin loadings 0.79–0.83.

4 Discussion

Principal finding. Sparse mFACEs with Chiou normalization jointly applied to eleven periprandial analyte-forms, paired with the PTP → IEP Type I–V framework presented in two parallel tracks (without and with glycemic subclassification), reveals six cohort-specific multi-hormonal signatures that are statistically robust under two orthogonal classification routes. Both routes converge on the same cohort extremes — reference minimum dysphysiological prevalence versus Obesity_T2DM/RYGBP/T2DM dominance — while preserving interpretable rank discordance in the rails' internal orderings (Spearman $\rho = 0.50$). The moderate concordance demonstrates that the two frameworks capture complementary rather than redundant information: joint Pillai F measures multivariate distance from reference; IEP Type IV.II prevalence measures per-analyte dysphysiology prevalence. Reporting both strengthens the scientific claim by triangulating cohort signatures from orthogonal methodological angles.

Conceptual contribution: physiologic periprandial responsiveness. The replacement of the conventional construct *secretory capacity* by *physiologic periprandial responsiveness* throughout the framework is more than terminological substitution. *Secretory capacity* is operationally measured by maximal-stimulation tests and reflects the biosynthetic pipeline (ER folding, hormone storage, granule biogenesis); it cannot describe the inhibition phase of a hormone like ghrelin, which is biologically defined by suppressing under satiety, nor can it capture the bidirectional FPCA-decomposed dynamics that this framework analyzes. *Physiologic periprandial responsiveness* unifies fasting, postprandial, secretory, and inhibitory phases under a single multi-analyte multi-state construct that is mathematically tractable via FPCA on multi-hormone trajectories. The neologism formalizes a distinction that the literature has long needed but has not had a precise term to express. The framework's evaluation of fasting hormones (ghrelin, glucagon) during their dominant (fasting) phase and of postprandial hormones (GIP, GLP-1, insulin, PYY) during their dominant (postprandial) phase, all anchored to the basal value, captures the entirety of the enteropancreatic responsive system rather than a single-direction projection of it.

Post-bariatric signature. RYGBP combines 37 % Altered + 12 % Enhanced in the joint rail and 76 % Type IV.II in the IEP rail (62.8 % IV.II.c when stratified by glycemia), with the most pronounced multi-axis remodeling and the largest insulin excursion vs. reference ($\Delta \approx +665$ pmol/L). This reproduces the exaggerated GLP-1 response documented mechanistically by Jørgensen and colleagues[14] via Exendin-9 blockade and the multi-hormone choreographic shift described by Madsbad, Dirksen and Holst.[13] SG shows an operationally distinguishable profile — the most pronounced ghrelin suppression across all

cohorts (sup-t band $|\Delta| = 143$ pmol/L, 100 % grid significant), consistent with the fundectomy mechanism of Ryan and colleagues[15] — but does not match RYGBP's combination of dysphysiology with incretin amplification; the distinction supports the pre-registered stance against pooling surgical modalities (v10.0 §3). Recovered ghrelin PTP in long-term post-RYGBP patients with sustained weight loss, baseline elevation, and significant postprandial suppression is consistent with the multi-axis recovery that the framework can capture only because it includes the basal value.

Diabetic and convergent signatures. T2DM without obesity shows 28 % Impaired + 9 % Blunted in the joint rail and 37 % Type III.II (infra-physiological moderate-severe) in the IEP rail without subclassification — a signature of dominant incretin secretion deficiency compatible with the GIP paradox update of Nauck and Müller.[18] The framework's inclusion of the basal (fasting) value as the first coordinate of the per-analyte classification vector permits, in principle, detection of fasting-GIP elevations preceding overt islet dysfunction; this is consistent with prior reports of altered fasting GIP in dysglycemic states[20,52] but is offered here as a hypothesis-generating observation that would require dedicated prediabetic / IGT cohorts (not represented in the current corpus) for confirmatory testing. Obesity_T2DM presents a convergent signature with Type IV.II.c dominance at 56 % — partially synergistic dysphysiology beyond the sum of individual cohort patterns, justifying the pre-registered discordance-escalation rule applied specifically to this arm (v10.0 §2.14.3). Pattern Type II as conceptual therapeutic target — supra-physiological coherent enteropancreatic responsiveness with preserved glycemia — embodies the framework's mechanism-anchored diagnostic philosophy and aligns with the bariatric remission literature,[44,45] even though the present ecological corpus does not contain enough optimally-restored post-bariatric phenotypes to detect Type II empirically.

Glycemic subclassification: added value of the dual-track presentation. The two-track presentation (Types I–V without glycemic subclassification, then Types III/IV/V with subclassification) deepens information without sacrificing the framework's primary IEP claim. The sub-classified track shows that the dominant dysphysiological signature in metabolic-disease cohorts (RYGBP IV.II.c = 63 %, Obesity_T2DM IV.II.c = 56 %) is overwhelmingly coupled to dysglycemia, while a meaningful subset of post-surgical patients (RYGBP IV.II.a = 10 %, SG IV.II.a = 8 %) shows dysphysiology with preserved glycemia — a phenotype potentially associated with post-bariatric reactive hypoglycemia or with optimal therapeutic restoration. This subphenotyping has prognostic and stratification implications that the unsubclassified Type I–V cannot capture; conversely, the unsubclassified track preserves the integrity of the enteropancreatic pattern claim independently of glycemic outcome — the framework's foundational claim that the integrated enteropancreatic pattern is biologically meaningful per se. Reporting both tracks reflects the pre-registered scope of this study and acknowledges that within-cohort heterogeneity across glycemic subtypes is an expected feature given the multidimensional variables that modulate physiologic periprandial responsiveness.

Methodological contributions. Three contributions. (i) Chiou normalization within joint mFACEs is

operationally indispensable: without it, PC1–PC2 are dominated by ghrelin and insulin absolute variance and the incretin axis vanishes (loading 0.000); after explicit Chiou scaling, $K = 14$ components are retained at FVE 0.90 and PC1 concentrates 81.7 % of its squared loading on incretins. (ii) Dual classification (joint-mFPC + per-analyte PTP/IEP) triangulates signatures by orthogonal methodologies. (iii) Dual-track IEP presentation (with/without glycemic subclassification) deepens information layering for diagnostic and stratification applications.

Limitations. Five material limitations bound interpretation. *First*, the *master data table* reports only cohort means; trajectory variance is inferred from literature-based CV priors, with the $CV \times \{0.75, 1.25\}$ sensitivity bounding the propagated effect on the primary conclusions. *Second*, inference is ecological pseudo-IPD (pre-registration v10.0 §4.2); $N = 2750$ saturates permutation p-values at the minimum $B^{-1} \approx 2 \times 10^{-4}$, so F magnitudes are reported as comparative cohort-separation metrics rather than as classical small-sample test statistics. *Third*, the per-analyte PTP framework applied across 10 non-glucose analytes with ±P5/P95 thresholds assigns a residual fraction of reference subjects to Type IV by multiple-testing arithmetic; we apply the framework's pathophysiological filter (PTP/IEP framework v1.0 §3.7; Supplementary File S4), which mitigates but does not eliminate this arithmetic floor. *Fourth*, dynamic enteropancreatic secretion captured by FPCA does not always correspond proportionally to biological action capacity — a fundamental limitation acknowledged in the framework's foundation that the diagnostic value of dynamics versus capacity differs across analytes and contexts (e.g., glucagon hyperglycemia driver in T2DM vs. glucagon protector against post-bariatric hypoglycemia: same secretory dynamics, opposite biological actions). *Fifth*, multi-cohort heterogeneity across protocols (caloric protocol, ingestion order, macronutrient proportions, chronobiology of testing, sex/age distributions, prior intake history, intervention type, sample handling) introduces substantial variability that this framework explicitly acknowledges as expected. The framework's contribution is the development of a multi-level multi-dimensional standardization tool tolerant to this heterogeneity, rather than its elimination.

Clinical implications. The framework operationalizes a quantitative basis for *physiologic periprandial responsiveness* as a multi-analyte construct distinct from static *secretory capacity*. The RYGBP signature (IEP Type IV.II.c with Enhanced incretin component) is operationally distinguishable from the T2DM pure signature (Type III.II infra-physiological) and from the Obesity_T2DM convergent signature (Type IV.II.c dominant). This suggests immediate clinical application: functional baseline stratification of bariatric surgery candidates by PTP/IEP pre-operative signature could identify candidates more likely to achieve Type II (supra-physiological coherent) post-surgery — the framework's ideal therapeutic restoration target — versus persisting in Type IV. Future expansion of the framework to anorexia, Prader-Willi syndrome, polycystic ovary syndrome, irritable bowel syndrome, and continuous glucose monitoring (CGM)-derived FPCA is enabled by the framework's design but is not undertaken here; nor is the secondary PTP analysis of within-subject pre-vs-post intervention pairs, which would require IPD that the present ecological corpus does not provide.

5 Data and Code Availability

Analysis code under CC-BY 4.0 at:

- Zenodo concept DOI: 10.5281/zenodo.19743544 (resolves to latest version)
- Zenodo v1.3 DOI (current — pre-journal-submission polish): 10.5281/zenodo.19758429
- Zenodo v1.2 DOI (historical — manuscript conceptual revision): 10.5281/zenodo.19750294
- Zenodo v1.0 DOI (historical — initial code snapshot): 10.5281/zenodo.19743545
- GitHub repository (current): sv8wmxnbp8-hash/EPP10 v1.3

Pre-registration with frozen analysis plan (2026-04-22) at:

- OSF registration DOI: 10.17605/OSF.IO/3CZRE
- OSF project: https://osf.io/tr469

Aggregate cohort-level data derived via the ETL are archived alongside the code (`hormones_long_tidy.csv`). The original *master data table* is retained under controlled access consistent with ICMJE data-sharing expectations; source-study reference list is preserved in the Verification Appendix S3.

6 Reporting Guidelines and Declarations

Reporting guideline. Reported in accordance with STROBE cohort[46] — completed checklist in Supplementary File S1. TRIPOD+AI[47] declared not applicable (descriptive-inferential mFPCA analysis, not a multivariable clinical prediction model).

Ethics and consent. Not applicable (ecological meta-analysis of publicly available aggregate data; no primary data collection).

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Author contributions (CRediT). Héctor Manuel Virgen Ayala (sole author): Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, review and editing, Visualization, Project administration.

AI/LLM disclosure. Analytic pipeline design, R code implementation, and manuscript drafting were supported by Claude (Anthropic, Opus 4.7, April 2026). All statistical outputs, pre-registration decisions, and scientific claims were validated by the author against the v10.0 master analysis plan (OSF

pre-registration) and the PTP/IEP framework v1.0 reference document (Supplementary File S4). AI is disclosed per openRxiv/ICMJE guidance; AI is not listed as an author.

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7 References

References are numbered in order of first appearance in the text. Numbers 1–48 are methodological, epidemiological, and mechanistic citations woven through the Introduction, Methods, Discussion and Reporting-Guidelines sections. Numbers 49–71 are the 23 primary source studies whose published aggregate data populated the *master data table* (71 Author × Cohort tuples; see Methods §2.1 and Supplementary S3 Verification Appendix). Formatting follows ICMJE/Vancouver recommendations for manuscripts submitted to biomedical journals.

7.1 Methodological, epidemiological, and mechanistic references

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Note: Reference 23 (Eli Lilly TRIUMPH-4) is a press release; the corresponding peer-reviewed publication is pending. Reference 24 (Garvey et al. REDEFINE 1) was upgraded from an early press-release citation to the peer-reviewed NEJM publication (DOI 10.1056/NEJMoa2502081; PMID 40544433). Reference 35 (Li, Xiao, Luo 2020 — Stat 9(1):e245) was corrected from an earlier draft citation pointing to a different Li et al. 2021 paper. Reference 45 (Bojsen-Møller et al. 2014 — Diabetes 63(5):1725–1737) author list verified against PubMed (PMID 24241533). Reference 52 (Di Giuseppe et al. 2025 — JCEM 111(1):e240–e246) title and pagination verified against the published JCEM record (DOI 10.1210/clinem/dgaf210). Source-study DOIs 49–71 correspond to the `Reference` *column of* `hormones_long_tidy.csv` *archived at Zenodo v1.2 (10.5281/zenodo.19750294); see Supplementary S3 for the full cohort-study mapping.*

8 Tables

Table 1. Cohort and hormone coverage.

Cohort v10.0	Source labels	Arms	Obs (long)	Pseudo-subj. N
`no_obese_without_T2DM`	1	13	692	650
Obesity	17	14	777	700
T2DM	2	4	280	200
Obesity_T2DM	10	7	528	350
SG	7	7	264	350
RYGBP	10	10	376	500
Total	47	55	2917	2750

Hormone	n obs	CV prior	Unit	Biological role
glucose	508	0.12	mmol/L	regulation signal
insulin	460	0.20	pmol/L	postprandial
GIP_total	380	0.30	pmol/L	postprandial
GLP1_active	315	0.40	pmol/L	postprandial
GLP1_total	300	0.35	pmol/L	postprandial

Hormone	n obs	CV prior	Unit	Biological role
PYY_total	215	0.35	pmol/L	postprandial
ghrelin_acyl	199	0.45	pmol/L	fasting
ghrelin_total	192	0.40	pmol/L	fasting
PYY_3_36	152	0.40	pmol/L	postprandial
glucagon	100	0.30	pmol/L	fasting
GIP_active	96	0.35	pmol/L	postprandial

Table 2. Joint-mFPC 6-class classification prevalence per cohort (% median [Q1–Q3] across B = 50 pipeline bootstrap re-fits). Modal class per cohort in bold.

Cohort	Preservado	Imp. lim.	Impaired	Blunted	Enhanced	Altered
`no_obese_without_T2DM`	**41.7** [39.5–42.8]	29.5 [27.5–31.3]	16.2 [14.8–17.9]	1.6 [0.9–2.1]	1.7 [1.1–2.8]	8.8 [7.3–11.2]
Obesity	33.9 [31.9–36.6]	29.8 [28.2–31.2]	22.2 [20.5–23.8]	3.7 [1.3–6.5]	2.1 [1.2–5.0]	6.0 [3.9–9.5]
T2DM	20.2 [18.5–24.0]	24.2 [21.1–27.0]	27.8 [25.1–30.0]	9.0 [7.0–15.8]	7.5 [1.9–12.9]	13.8 [10.5–19.2]
Obesity_T2DM	19.4 [16.6–20.8]	23.9 [22.0–26.2]	27.6 [25.2–29.6]	10.9 [0.9–14.9]	4.9 [0.6–11.1]	15.0 [11.7–19.2]
SG	25.4 [22.1–29.5]	24.7 [23.2–27.1]	22.6 [21.7–25.0]	2.3 [0.6–8.9]	4.9 [0.6–13.0]	14.7 [10.4–20.6]
RYGBP	9.2 [7.6–11.4]	15.4 [13.6–16.4]	20.3 [18.8–22.0]	2.4 [0.4–16.4]	12.4 [0.5–21.0]	37.1 [34.4–40.2]

Table 3. Formal inference. Panel A omnibus + pairwise Pillai. Panel B top-three per-PC F. Panel C significant-PC coverage (FDR < 0.05).

Panel A — Pillai multivariate contrasts. F values are reported as cohort-separation magnitudes (see Methods §2.9 and Results §3.4); permutation p-values are saturated against the minimum permutation grain $B^{-1} \approx 2 \times 10^{-4}$ for every contrast under pseudo-IPD inflation. Pipeline-stage bootstrap (B = 50) confirmed cohort-ordering invariance (T2DM > RYGBP > Obesity_T2DM > SG > Obesity) in 50/50 reps.

Contrast	Pillai F	p_perm	p_adj_BH
Omnibus (6 cohorts)	47.6	$< B^{-1}$	$< B^{-1}$
T2DM vs. ref	108.3	$< B^{-1}$	$< B^{-1}$
RYGBP vs. ref	86.8	$< B^{-1}$	$< B^{-1}$
Obesity_T2DM vs. ref	59.3	$< B^{-1}$	$< B^{-1}$
SG vs. ref	34.1	$< B^{-1}$	$< B^{-1}$
Obesity vs. ref	22.5	$< B^{-1}$	$< B^{-1}$

Panel B — Dominant per-PC F statistics

Contrast	Dominant PC	Secondary PC	Tertiary PC
RYGBP vs. ref	PC1 (F = 496)	PC7 (F = 275)	PC4 (F = 244)
T2DM vs. ref	PC7 (F = 370)	PC1 (F = 153)	PC3 (F = 104)
Obesity_T2DM vs. ref	PC1 (F = 273)	PC7 (F = 223)	PC2 (F = 147)
SG vs. ref	PC1 (F = 139)	PC2 (F = 88)	PC11 (F = 83)
Obesity vs. ref	PC1 (F = 102)	PC2 (F = 97)	PC7 (F = 36)

Panel C — Coverage of significant PCs

Contrast	PCs sig. / 14
Omnibus	14 / 14
Obesity_T2DM vs. ref	11 / 14
T2DM vs. ref	10 / 14
RYGBP vs. ref	9 / 14
Obesity vs. ref	8 / 14
SG vs. ref	8 / 14

Table 4A. PTP/IEP Type I–V distribution per cohort, without glycemic subclassification, within the classifiable subset (1600 / 2750 = 58 %). Modal Type per cohort in bold. Final column reports IV.II prevalence relative to reference (cohort − reference, percentage points) to control for the per-analyte arithmetic floor described in §2.10.6.

Cohort	I.I	III.I	III.II	IV.I	IV.II	V.I	V.II	Δ IV.II vs. ref (pp)
no_obese_without_T2DM (ref)	6.8	2.8	12.5	47.8	18.8	1.2	0.2	—
Obesity	2.9	3.1	4.3	44.6	45.1	0	0	+26.3
T2DM	0	0	37.0	34.5	28.5	0	0	+9.7
Obesity_T2DM	0	3.7	6.7	33.7	56.0	0	0	+37.2
SG	0	2.0	0	43.0	55.0	0	0	+36.2
RYGBP	0	0.4	0.8	22.4	76.4	0	0	+57.6

Table 4B. PTP/IEP Type I–V with glycemic subclassification a/b/c restricted to Types III, IV, V (Patterns I and II by definition imply preserved glycemia and are not subclassified). Modal subtype per cohort in bold. Subtype a = preserved/recovered; b = lower-tail glucose; c = upper-tail/dysglycemic.

Cohort	I.I	III.I.a	III.I.b	III.I.c	III.II.a	III.II.b	III.II.c	IV.I.a	IV.I.b	IV.I.c	IV.II.a	IV.II.b	IV.II.c	V.I.a	V.II.a
ref	6.8	2.8	2.0	8.0	3.2	2.8	6.5	15.8	9.8	22.2	7.8	2.2	8.8	1.2	0.2
Obesity	2.9	0	0	3.1	0	0	4.3	2.3	0.3	42.0	1.1	7.4	36.6	0	0
T2DM	0	0	0	0	0	0	37.0	0	0	34.5	0	0	28.5	0	0
Obesity_T2DM	0	0	0	3.7	0	0	6.7	0	0	33.7	0	0	56.0	0	0
SG	0	0	0	2.0	0	0	0	4.0	0	39.0	8.0	0	47.0	0	0
RYGBP	0	0	0	0.4	0	0	0.4	3.6	0	18.8	10.4	3.2	62.8	0	0

9 Figures

Figure 1. Cohort composition and data-flow characteristics.

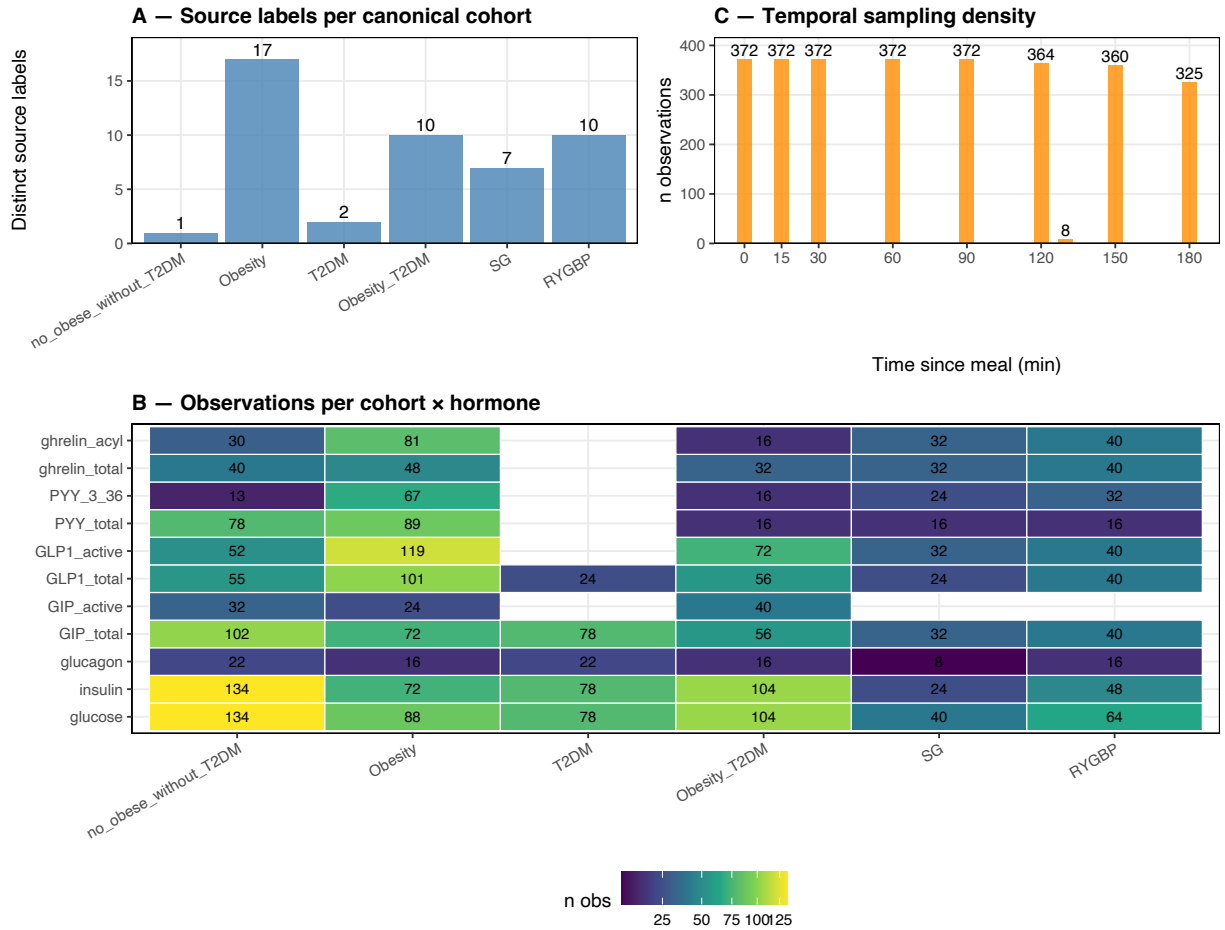

Figure 1: Figure 1

Figure 2. Classification prevalence (joint 6-class, IEP Type I–V), Pillai F, and cross-framework concordance.

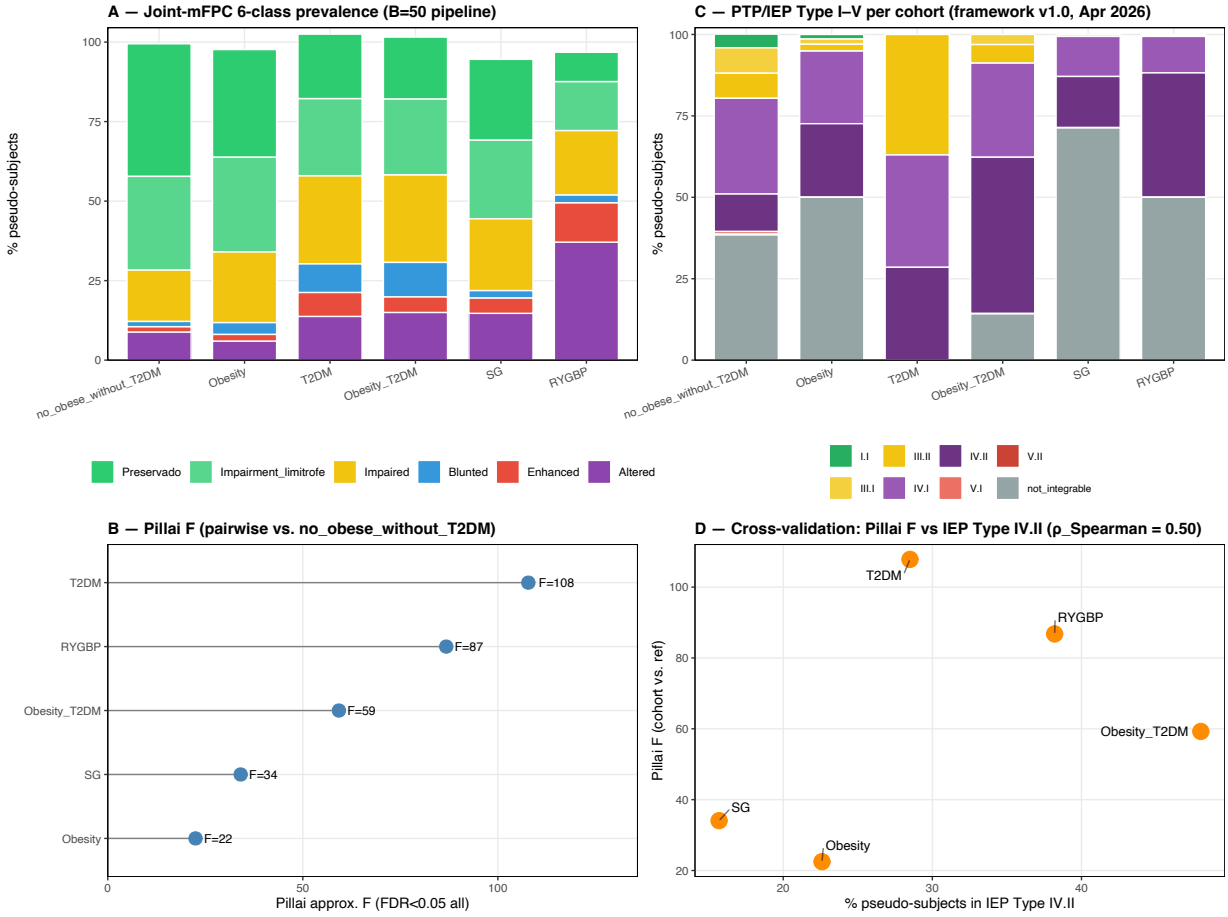

Figure 2: Figure 2

Figure 3. Trajectory differences with 95 % simultaneous sup-t bands.

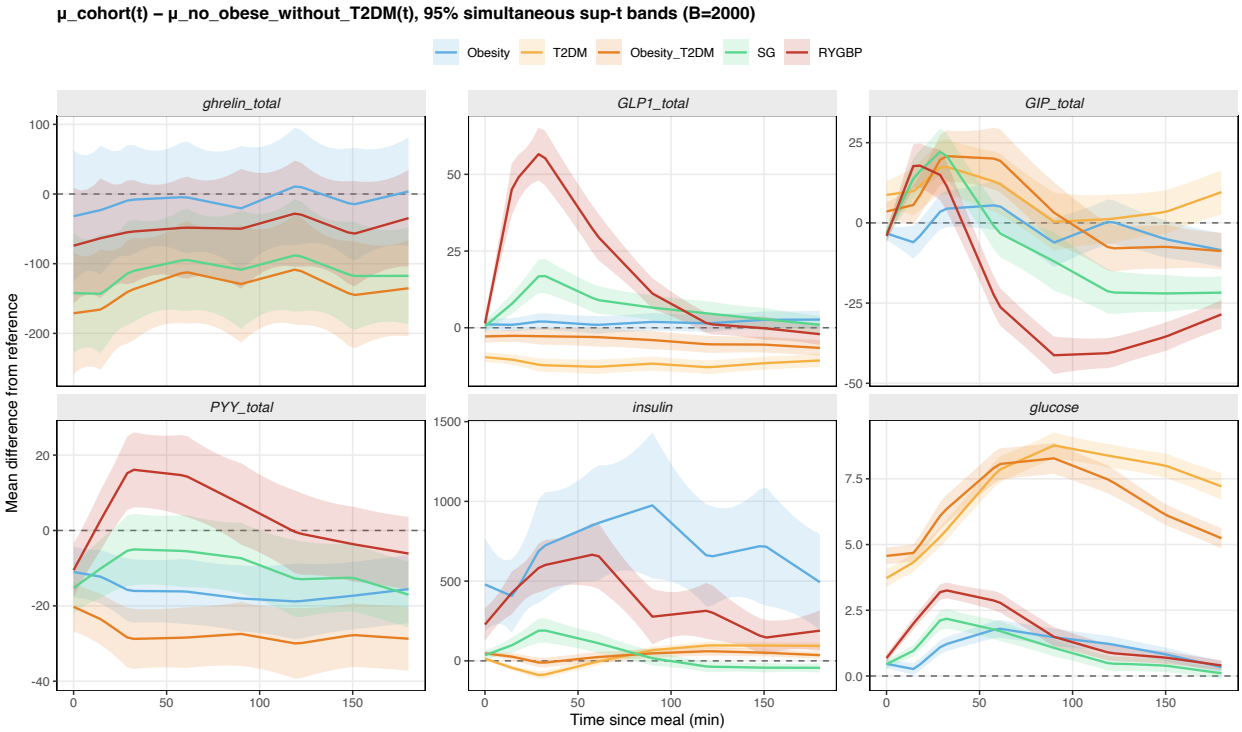

Figure 3: Figure 3